

Description of the Real Data Sets

Women's Mobility and the MathExam14W mathematics test data

This note documents the two real data sets used in the real data analysis of the thesis: the **Women's Mobility** data and the **MathExam14W** mathematics test data. Both data sets are samples of binomial counts with a **fixed number of Bernoulli trials**: each subject responds to m binary items, and the recorded observation X is the number of successes, so that X takes values in $\{0, 1, \dots, m\}$ with the same m for every subject. For each data set we report the source and the way it is accessed in R, the background of the study, the sample size together with the main features of the data, and the definition of every variable, distinguishing the response variable used for model fitting from the available covariates.

1 Women's Mobility data

1.1 Source and availability

Primary source	A rural subsample of the Bangladesh Fertility Survey of 1989, conducted by the National Institute of Population Research and Training (NIPORT), Dhaka (Huq and Cleland, 1990). The data were subsequently compiled and analysed by Bartholomew, Steel, Moustaki and Galbraith (2002).
R package and call	<i>The data set is distributed with the ltm package for latent variable modelling (Rizopoulos, 2006) and is loaded by</i> library(ltm) data(Mobility) <i>The resulting object is a data frame with 8445 rows and 8 columns, named Item 1 to Item 8.</i>

1.2 Background

The survey measures the social freedom of women in rural Bangladesh outside the household. Each respondent was asked whether she was able to perform, *alone*, eight activities of daily life, namely going to any part of the village, town or city; going outside the village, town or city; talking to an unknown man; going to a cinema or cultural show; going shopping; going to a cooperative, mothers' club or other club; attending a political meeting; and going to a health centre or hospital. The activities range from local movement to public participation, so their difficulty and their social acceptability differ substantially, and the degree of freedom varies considerably across women.

1.3 Sample size and data characteristics

Sample size	$n = 8445$ women; each woman answered all $m = 8$ items, with no missing values.
Binomial structure	Treating the eight items as Bernoulli trials, the response is $X = \text{number of activities that the woman may perform alone}$, X in $\{0, 1, \dots, 8\}$, and $X p \sim \text{Binomial}(8, p)$, where p denotes the woman-specific level of mobility.
Overall level	Averaged over women and items, the proportion of activities that may be performed alone is 0.3187.
Distributional features	The score distribution is strongly over-dispersed, left-skewed and non-unimodal . The mode is attained at $X = 2$ (28.1 per cent), while a clearly separated secondary mode appears at $X = 8$, i.e. 246 women (2.9 per cent) may perform all eight activities alone. The distribution therefore mixes a large restricted subpopulation with a small fully unrestricted one, a structure that no unimodal prior on p can represent.

Table 1. Observed distribution of the Mobility score X (number of activities that may be performed alone).

X	0	1	2	3	4	5	6	7	8
Frequency	829	1617	2369	1533	1126	389	193	143	246
Percentage	9.8	19.1	28.1	18.2	13.3	4.6	2.3	1.7	2.9

1.4 Variables

The raw data consist of eight binary item indicators. The variable used for model fitting is the total score obtained by summing these indicators.

Variable	Values	Description	Role
Item 1	0 / 1	Going to any part of the village, town or city	Item
Item 2	0 / 1	Going outside the village, town or city	Item
Item 3	0 / 1	Talking to an unknown man	Item
Item 4	0 / 1	Going to a cinema or cultural show	Item
Item 5	0 / 1	Going shopping	Item
Item 6	0 / 1	Going to a cooperative, mothers' club or other club	Item
Item 7	0 / 1	Attending a political meeting	Item
Item 8	0 / 1	Going to a health centre or hospital	Item
score	0 - 8	Number of activities that may be performed alone , obtained as <i>rowSums(Mobility)</i> ; this is the response X modelled by $\text{Bern-Bino}(8, K, \lambda)$	Response

Note. For every item, 1 = the activity may be performed alone and 0 = it may not. The eight items are not equally restrictive: the corresponding proportions of 'yes' answers are 79.9, 31.4, 75.2, 36.5, 6.9, 11.1, 5.3 and 8.7 per cent, respectively. The data set contains no covariates.

2 MathExam14W mathematics test data

2.1 Source and availability

Primary source	Individual results of the written final examination and of the accompanying online exercises of the course 'Mathematics 101' for first-year business and economics students at the University of Innsbruck, Austria. The data were collected by the Department of Statistics of the University of Innsbruck and are documented by Zeileis, Umlauf and Leisch (2014).
R package and call	<i>The data set is distributed with the psychotools package and is loaded by library(psychotools) data("MathExam14W", package = "psychotools") The resulting object is a data frame with 729 rows and 9 columns.</i>

2.2 Background

The course assessment combined biweekly online exercises with a written final examination. The online component consisted of 26 numerical exercises administered through the OpenOLAT platform, while the written examination consisted of 13 single-choice items, each offering five answer alternatives; an incorrect answer was penalised by 25 per cent of the item value in order to discourage random guessing. Of the 941 students registered for the course, 729 took the examination. Because of the large number of participants the students were divided into two groups, which received partially different versions of items 1, 5, 6, 7, 8, 9, 11 and 12; the versions differed in the story or the numerical set-up but not in the mathematical skills required to solve them.

2.3 Sample size and data characteristics

Sample size	$n = 729$ students; each student answered all $m = 13$ items, with no missing values.
Binomial structure	Treating the 13 items as Bernoulli trials, the response is $X =$ number of items solved correctly, X in $\{0, 1, \dots, 13\}$, and $X p \sim \text{Binomial}(13, p)$, where p denotes the student-specific ability.
Distributional features	The distribution of the number of solved items is unimodal and slightly right-skewed, with the mode at $X = 7$ and $X = 8$ (104 and 98 students, respectively). Both tails are long: 9 students solved no item at all and 32 students solved all 13 items. The item difficulties differ considerably, the proportion of correct answers ranging from 17.4 to 78.5 per cent, which indicates substantial heterogeneity in ability and makes a simple binomial model inadequate.
Grouping and pass mark	Group 1 contains 334 students and group 2 contains 395 students; solving at least 8 of the 13 items is conventionally required to pass the course.

Table 2. Observed distribution of the MathExam14W score X (number of items solved correctly).

X	0	1	2	3	4	5	6	7	8	9	10	11	12	13
Frequency	9	16	24	33	50	65	71	104	98	81	67	50	29	32
Percentage	1.2	2.2	3.3	4.5	6.9	8.9	9.7	14.3	13.4	11.1	9.2	6.9	4.0	4.4

2.4 Variables

The data set contains nine variables: *solved* and *credits* are item-by-item response matrices, *nsolved* is the response used here, and the remaining variables are covariates. The regression analysis of the thesis uses the response *nsolved* together with four covariates, namely *tests*, *gender*, *study* and *semester*.

Variable	Type / values	Description	Role
<i>solved</i>	matrix, 13 columns of 0/1	Item-by-item response, 1 = solved correctly and 0 = otherwise	Source of the response
<i>credits</i>	matrix, 13 columns of 0/1/2	Item-by-item credit, 2 = correct, 1 = incorrect, 0 = not attempted	Not used
<i>nsolved</i>	integer, 0 - 13	Number of items solved correctly = the response X	Response
<i>tests</i>	integer, 9 - 26	Number of online test exercises solved correctly before the examination , a measure of study effort	Covariate
<i>gender</i>	female / male	Gender (326 female and 403 male students)	Covariate

Variable	Type / values	Description	Role
study	155 / 571	Type of degree programme: 155 = four-year diploma programme (146 students) and 571 = three-year bachelor programme (583 students)	Covariate
semester	integer, 1 - 21	Number of semesters enrolled in the programme (median 1, mean 3.11)	Covariate
attempt	factor, levels 1 - 5	Number of times the course or examination has been attempted (431, 52, 121, 113 and 12 students for levels 1 to 5)	Not used
group	factor, levels 1 and 2	Examination group; the two groups received partially different item versions	Not used

Note. Although *attempt* is also a covariate, its five levels are highly unbalanced because the vast majority of the students sat the examination for the first time; including it as a set of dummy variables adds noise and reduces the out-of-sample performance, so it is not used in the regression model. The variable *group* reflects the administrative division of the examination rather than a characteristic of the student and is therefore not used either.

3 Concluding remarks

Both data sets satisfy the requirements of the model considered in this thesis: the observations are numbers of successes in repeated Bernoulli trials and therefore take values in the finite discrete set $\{0, 1, \dots, m\}$; the number of trials is fixed at $m = 8$ for the Mobility data and at $m = 13$ for the MathExam14W data; and the sample sizes of 8445 and 729 observations are sufficient to estimate the additional weights of a Bernstein prior. Moreover, both score distributions display features that a unimodal parametric prior cannot reproduce, namely the pronounced secondary mode at $X = 8$ in the Mobility data, and the wide, long-tailed shape of the score distribution of the examination data. The two data sets consequently serve complementary purposes in the real data analysis:

- **Women's Mobility** contains no covariates and is used for the comparison of the fitted distributions, namely the Binomial, the Beta-binomial, the logit-normal binomial, the Kumaraswamy binomial and the proposed Bern-Bino distribution.
- **MathExam14W** contains four usable covariates and is used for the comparison of the regression models of Section 2.5, namely logistic regression, beta-binomial regression and Bern-Bino regression.

References

- [1] Bartholomew, D. J., Steel, F., Moustaki, I. and Galbraith, J. I. (2002). *The Analysis and Interpretation of Multivariate Data for Social Scientists*. London: Chapman and Hall.
- [2] Huq, N. and Cleland, J. (1990). *Bangladesh Fertility Survey, 1989*. Dhaka: National Institute of Population Research and Training (NIPORT).
- [3] Rizopoulos, D. (2006). ltm: An R package for latent variable modelling and item response theory analyses. *Journal of Statistical Software*, 17(5), 1-25.
- [4] Zeileis, A., Umlauf, N. and Leisch, F. (2014). Flexible generation of e-learning exams in R: Moodle quizzes, OLAT assessments, and beyond. *Journal of Statistical Software*, 58(1), 1-36.